SolvePath

Layered architecture

A deterministic workflow harness with bounded LLM stages. No model chooses a tool, route, verdict, hint level or state transition.

- **solvepath-mcp** — Verification & retrieval (MCP)
- **LLM adapter** — Schema-bound model calls
- **solvepath-safety** — A2A safety verdicts
- **Neon Postgres** — Durable state & objects
- **Langfuse** — Telemetry / in-process

1 Presentation & client

apps/web

UI
React 18 SPA
Vite + TypeScript · renders only tutor.controls

UI_STATE
State & cache
TanStack Query + KaTeX

CLERK_SDK
Clerk auth SDK
Browser token

HTTPS JSON + Clerk JWT

JWKS proof

2 Edge & security ingress

apps/api · L1 Edge

PROXY
Same-origin /api/* proxy
Vercel Edge rewrite, no CORS

→

AUTH_GATE
Auth & token scope gate
Clerk JWKS cached ≤ 1 h · role from Postgres

→

SEC_MIDDLEWARE
Security middleware
Rate limiter · idempotency · PII scrubber

Validated request

3 Routing & application

apps/api · L2 / L3

FASTAPI_ROUTERS
FastAPI routers
Student · educator · admin · jobs

→

ORCHESTRATOR
Tutor Orchestrator
run_turn hub & session lease manager
→ **Neon Postgres** — Unit of work & audit
→ **Langfuse** — OTLP telemetry

SVC_MANAGERS
Application services
Sessions · transfers · job runners
→ **Neon Postgres** — Read / write
→ **Vercel Blob** — Diagram retrieval

run_turn → SequentialAgent

4 Agent workflow

apps/api · L5 Google ADK

DETERMINISTIC SEQUENTIALAGENT PIPELINE

AG-01
Problem Context Agent
Load · scrub · normalise · Input Screen

→

AG-02
Math Verifier Agent
→ **solvepath-mcp** — HTTP + service token
→ **Math normaliser** — In-process fallback

→

AG-03
Misconception Matcher

→

AG-04
Retrieval Gate
→ **solvepath-mcp** — HTTP search

→

AG-05
StudentState Estimator

→

AG-06
Misconception Classifier
LLM
→ **LLM adapter** — Schema bound

→

AG-07
HintPolicy Engine
Level · probe · prompt type

AG-07 hands off ↓

LOOPAGENT · MAX 2 ITERATIONS · CANDIDATE → CHECK → GUARD

AG-08
Socratic Dialogue Agent
LLM · phrases only what was selected
→ **LLM adapter** — Schema bound

→

AG-09
Deterministic Leak Check
A hit always beats guard allow

→

AG-10
Safety Guard
A2A client
→ **solvepath-safety** — A2A HTTP

Evaluates against

Evaluates against

5 Pure deterministic engines

apps/api · L4 · no I/O, no model calls

FSM
Table-driven FSM
transitions.apply

MATH_ENGINE
Math normaliser
SymPy & mpmath

HINT_RULES
Pedagogical hint engine
L1–L5 rules

LEAK_RULES
Safety & leak check rules
Deterministic

Unit of work & audit

6 Persistence & storage

Neon & Blob · L6

NEON_DB
Neon PostgreSQL + pgvector
Domain tables · skill graph · audit (insert-only)

VERCEL_BLOB
Vercel Blob storage
Diagram SVGs · item assets

7 Satellites & external integration

L7 adapters · service tokens

MCP_SERVER
solvepath-mcp
Math verifier & exemplar search
← AG-02 · verify_expression
AG-04 · search_exemplars
→ **Neon Postgres** — Read-only SQL

A2A_SAFETY
solvepath-safety
Small-LLM safety guard service
← AG-10 · allow / block / revise / escalate

LLM_PROVIDER
LLM adapter
Gemini / OpenRouter / vLLM
← AG-06 · classifier
AG-08 · dialogue

LANGFUSE
Langfuse
OTLP spans & prompt versioning
← ORCHESTRATOR · pseudonymous ids only

Downward-only dependencies. L4 imports nothing from L5–L7. **Fail closed.** Every dependency failure degrades to a deterministic, mathematics-free fallback. **Source:** uploads/mermaid-diagram.svg · DESIGN.md §1